

CISC 365 Algorithms I Course Syllabus

Revised September 9, 2026

Course Logistics

Offering: Fall 2026
 Modality: face to face
 Prerequisites: Registration in a SoC plan and a C- or higher in CISC 203, 204, and 235
 Lecture dates: September 8 ~ December 8, 2026
 Lecture time: Mondays 11:30am-12:20pm
 Tuesdays 13:30-14:20pm
 Thursdays 12:30-13:20pm
 Lecture location: Stirling B
 Instructors: Prof. Yuanzhu Chen
 Prof. Xu Wang
 TAs: Felix Xing, James Song, Mohamed Harmanani
 Office hours: Instructor - 2:30-3:20pm, Tuesdays
 in Goodwin 536 and
 on MS Teams (channel "Office hour with Instructor")
 TA - 5:00-6:00pm, Thursdays,
 in Goodwin 536 and
 on MS Teams (channel "Office hour with TA")

Out-of-classroom Communications

Online: MS Teams
 Emails: cisc365@cs.queensu.ca, yuanzhu.chen@queensu.ca
 - Use the Queen's email addresses above (not OnQ)
 - Include "CISC365" in email subject, and name and student number in email body
 In person: by appointment

Indigenous Land Acknowledgement

Queen's University is situated on traditional Anishinaabe and Haudenosaunee Territory.

Course Learning Outcomes:

- Possess a strong understanding of computational complexity, up to and including polynomial time reducibility; class P and NP; NP-Completeness; proofs of NP-Completeness
- Be able to recognize classical NP-Complete problems
- Be able to prove that new problems are NP-Complete using polynomial time reductions from known NP-Complete problems
- Apply key algorithm paradigms, both in the abstract and through concrete examples: divide-and-conquer; greedy algorithms; dynamic programming; branch-and-bound

Topics:

1. Algorithm efficiency and problem complexity
2. Divide-and-conquer
3. Greedy strategies
4. Dynamic programming
5. Branch-and-bound

Assessments

- Term tests (× 5, in class) 20% each

Textbook

Introduction to Algorithms (4ed) by Cormen, Leiserson, Rivest, and Stein, MIT Press 2022.

Tentative Schedule

<i>Week</i>	<i>Topics (Chapters)</i>	<i>Dates</i>	<i>Evaluation</i>
1	1a: Algorithm complexity (Ch 2)		
2	1a: Algorithm complexity (Ch 3)		
3	2: Divide-and-conquer (Ch 4)	September 24	Test 1
4	2: Divide-and-conquer (Ch 4)		
5	3: Greedy strategy (Ch 15)	October 8	Test 2
6	Fall reading break		
7	3: Greedy strategy (Ch 15)		
8	4: Dynamic programming (Ch 14)	October 29	Test 3
9	4: Dynamic programming (Ch 14)		
10	5: Branch-and-bound	November 12	Test 4
11	5: Branch-and-bound		
12	1b: NP-completeness (Ch 34)		
13	1b: NP-completeness (Ch 34) review	December 1	Test 5

Policies and references

1. Academic Integrity

Academic Integrity is constituted by the six core fundamental values of honesty, trust, fairness, respect, responsibility, and courage (see www.academicintegrity.org). These values are central to the building, nurturing, and sustaining of an academic community in which all members of the community will thrive. Adherence to the values expressed through academic integrity forms a foundation for the "freedom of inquiry and exchange of ideas" essential to the intellectual life of the University (see the Senate Report on Principles and Priorities <http://www.queensu.ca/secretariat/policies/senate/report-principles-and-priorities>).

Students are responsible for familiarizing themselves with the regulations concerning academic integrity and for ensuring that their assignments conform to the principles of academic integrity. Information on academic integrity is available in the Arts and Science Calendar (see Academic Regulation 1 <https://www.queensu.ca/academic-calendar/arts-science/academic-regulations/>), on the Arts and Science website. Departures from academic integrity include plagiarism, use of unauthorized materials, facilitation, forgery and falsification, and are antithetical to the development of an academic community at Queen's. Given the seriousness of these matters, actions which contravene the regulation on academic integrity carry sanctions that can range from a warning or the loss of grades on an assignment to the failure of a course to a requirement to withdraw from the university.

The instructor reserves the right to give the student a failing course grade as a result of breaches of academic integrity deemed serious, including plagiarism of assignments.

2. Student Success Resources

- SASS online tutorial on academic integrity: a fundamental aspect of university (<https://sass.queensu.ca/resources/online/academic-integrity>)
- QSAS (Queen's Student Accessibility Services) Academic Accommodation for students with functional limitations or disabilities (<https://www.queensu.ca/studentwellness/accessibility-services/new-students>)
- Academic Consideration for students in extenuating circumstances (<https://www.queensu.ca/artsci/undergrad-students/student-services/academic-consideration>)

3. Equity, Diversity, and Inclusivity

Queen's University recognizes that the values of equity and diversity are vital to and in harmony with its educational mission and standards of excellence. It acknowledges that direct, indirect and systemic discrimination exists within our institutional structures, policies and practices and in our community. These take many forms and work to differentially advantage and disadvantage persons across social identities such as race, ethnicity, disability, gender identity, sexual orientation, faith and socioeconomic status, among other examples.

4. Copyright of Course Materials

The material on this website is copyrighted and is for the sole use of students registered in the course. The material on this website may be downloaded for a registered student's personal use but shall not be distributed or disseminated to anyone other than students registered in the course during the current term. Failure to abide by these conditions is a breach of copyright and may also constitute a breach of academic integrity under the University Senate's Academic Integrity Policy Statement.

5. Assessment Item Remarks

You may request a remark of your tests. All remark requests should be emailed to the instruction team's email (cisc365@cs.queensu.ca) within 5 calendar days after the grades are posted. Late requests will not be considered.

7. General Information

Please see <https://www.cs.queensu.ca/undergraduate/syllabus/> for the School's Common Syllabus Information.